MILITARY SPECIFICATION SHEET

ELECTRON TUBE, VOLTAGE REGULATOR

TYPE 8515

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Corona discharge

See figure 1

Mounting position: Any

Weight: 0.5 ounce nominal

ABSOLUTE RATINGS:

Parameter:	(A) Ez	(A) Eb	Ib	TA
Unit:	Vdc	Vdc	μAdc	°C
Maximum:	1,760	1,645	800	+85
Minimum:	---	1,550	20	-55
TEST CONDITIONS:	---	---	100	---

GENERAL:

Qualification - Required

Holding period (MIL-STD-1311) t = 7 days

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1201	Short and discontinuity detection		0.4	II	---	---	---	---
---	Holding period end points:	See note 6						
3347	Ionization voltage		0.65	II	Ez	---	1,760	Vdc
---	Change in voltage drop				ΔEb(1)	---	±15	Vdc
3335	Regulation				ΔEb	---	42	Vdc
---	Peak current	See note 5			---	---	950	μAdc
3347	Ionization voltage (total darkness)	See note 3	0.65	II	Ez	---	1,760	Vdc
3337	Voltage drop	Ebb/Ib = 100 ±2.0 μAdc	0.65	II	Eb	1,560	1,640	Vdc
3335	Regulation	Ib = 20 to 800 μAdc (see note 4)	0.65	II	ΔEb	---	42	Vdc
---	Peak current	See note 5	0.65	II	---	---	950	μAdc
	Quality conformance inspection, part 2							
1031	Vibration	No voltages; t = 5 minutes; 55 to 500 Hz; 10 G (max) (see note 1)	---	---	---	---	---	---
3305	Leakage current	Eb = 1,440 Vdc (min); T = 25° ±5°C; relative humidity = 90 +5%	---	---	Llb	---	0.5	μAdc
1026	Temperature range, operating	T = -55° ±5°C	---	---				
		Ionization voltage			Ez	---	1,760	Vdc
		Voltage drop			Eb	1,550	1,642	Vdc
		Regulation			ΔEb	---	42	Vdc
1026	Temperature range, operating	T = +85° ±5°C	---	---				
		Ionization voltage			Ez	---	1,760	Vdc
		Voltage drop			Eb	1,558	1,645	Vdc
		Regulation			ΔEb	---	47	Vdc
	Quality conformance inspection, part 3							
---	Life-test provisions	Group B; Ebb/Ib = 100 ±2 μAdc; t = 1,000 hours	---	---	---	---	---	---
---	Life-test end points:							
3337	Voltage drop		---	---	Eb	1,550	1,645	Vdc
3335	Regulation		---	---	ΔEb	---	47	Vdc
3347	Ionization voltage		---	---	Ez	---	1,760	Vdc
	Periodic-check test							
Ⓐ 1042	Shock, specified pulse	Test condition C (see note 2)	---	---	---	---	---	---

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NOTES:

1. Criterion for passing this test shall be compliance after test with the initial requirements for voltage drop and regulation.

2. Electrical samples shall not be used for this test. Ten additional samples shall be submitted for shock test only. Criterion for passing this test shall be compliance after test of at least 80 percent of the tubes with the initial requirements of voltage drop and regulation.

3. The tube under test shall not have been conducting for at least 30 minutes prior to test. During this test the regulator tube shall not be exposed to external sources of radiation and shall be shielded from light. No conditioning current is permitted.

4. Eb maximum shall not be exceeded anywhere within the regulated current range. In testing for regulation, current shall be varied continuously through the regulated range.

5. Peak current is the current maximum during a transient pulse which will not cause damage to the tube or cause tube to go into glow.

6. The holding period commences upon completion of the initial performance of the indicated tests. At the conclusion of the holding period the indicated tests will be performed again. Of the tubes failing the holding-period end-point test, those which fail with respect to voltage drop only, may be retested after an additional 60-day period. $\Delta Eb(1)$ is the change in voltage drop from beginning to end of holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Navy -
Air Force - 11, 17
DSA - ES

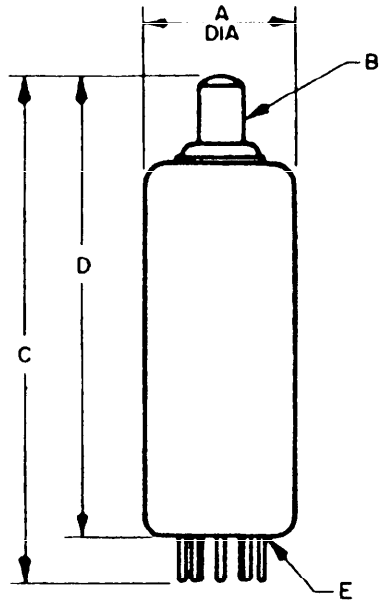
User activities:
Army - WC
Navy - AS, MC, CG, SH
Air Force -

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2464)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		.750 (19.05)
C		2.750 (69.85)
D		2.500 (63.50)
Reference dimensions		
B	CAP: C1-3 (EIA)	
E	BASE: E7-1 (EIA)	

Ⓐ FIGURE 1. Outline drawing of electron tube type 8515.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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